

Finite Sibling-Interface Descriptors for $\mathcal{ALCI}_{RCC5}$ (v2, Repaired)

Reinterpreted split-tree presentation, ambient root removed,
PP/PPI eventualities by reachability discharge

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Abstract

This is a repaired version of GPT-5.4’s sibling-interface note [1] (April 2026), which proves the soundness direction of the split-forest decidability programme for $\mathcal{ALCI}_{RCC5}$. GPT-5.5’s review [2] (May 2026) found that the v1 paper’s soundness chain (Thms 1.17–1.19) is intact, but identified two structural defects elsewhere in the document: (i) the split-tree presentation [1, Def 1.5] requires an *ambient root* above the evaluation point, which fails for formulas forcing infinite PP-chains (e.g. $C_{\uparrow} := \exists PP.\top \sqcap \forall PP.\exists PP.\top$); and (ii) the proof sketch of Thm 1.20 [1, Thm 1.20 sketch] extracts “least locally coherent support-closed tables” without constructing them and conflates witnesses across distinct occurrences when discharging the C5 triple-coherence axiom.

This v2 paper applies three small but conceptually significant repairs while preserving every soundness statement of v1 verbatim. First, the split-tree presentation is reinterpreted as a *generated* witness-cover presentation (parent function may self-loop at the root, so $C_{\uparrow}$ is represented by a single-state cycle on the parent function). Second, PP/PPI eventualities are discharged by *finite-graph reachability* on the quotient’s parent graph, replacing the witness-menu treatment of v1 line 367 which left transitive eventualities under-specified. Third, the Thm 1.20 proof sketch is dropped: the model-to-quotient direction is delegated to the explicit construction in the companion v2 completeness-extraction paper [6], which builds the support-closed mosaics that the v1 sketch only asserted.

The soundness chain valid finite quotient $\rightarrow$ disjunctive network $\rightarrow$ arc-consistency $\rightarrow$ König $\rightarrow$ canonical refinements $\rightarrow$ weak-EQ model $\rightarrow$ strong-EQ model (v1 Thms 1.17–1.19) is unchanged. Combined with v2 completeness extraction, this yields a clean decidability theorem for concept satisfiability in $\mathcal{ALCI}_{RCC5}$ under strong-EQ semantics, without any ω -acceptance machinery.

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1 What this v2 paper is and is not

1.1 Pointer to v1 for unchanged material

Every definition and theorem of GPT-5.4's v1 paper [1] is retained verbatim except where this note explicitly replaces or reinterprets it. Specifically, the following v1 material is unchanged and is cited rather than reproduced:

- v1 §2 (RCC5 conventions, types, modal depth, type-safe open labels, per-relation Need_R families).
- v1 §4 (the corrected status partition: *Core*, *Out*, *Front*, partition lemma).
- v1 §5 (rigid DR behaviour: uniform-DR lemma and the open cross-edge lemma).
- v1 §6 (status evolution; rank-0 colors; endpoint summaries; rank- k descriptors; local coherence axioms (C1)–(C5)).
- v1 §7 (rank- $(k+1)$ states, except for the Wit_k component, which is reinterpreted in Section 3 below; the finite-index Theorem 1.10 is unchanged).
- v1 §8 (valid finite rank- d quotient, unfolding $U(Q)$, disjunctive network of a finite prefix).
- v1 §9 (finite-prefix arc-consistency, Theorem 1.17).
- v1 §10 (prefix extension, canonical refinements, typed-EQ congruence, EQ-synchrony).
- v1 §11 (weak-EQ realisation, Theorem 1.18; strong-EQ realisation, Theorem 1.19).

Replaced or reinterpreted. The following v1 items are altered:

- v1 Def 1.5 (weak-EQ split-tree presentation, line 158–172): the requirement of an *ambient root* ϱ_T is dropped; tree edges are reinterpreted as *generated* witness-cover edges relative to a finite selected witness set, not as Hasse covers of an ambient PP-order; the parent function may carry a self-loop at the root to represent infinite PP-chains.
- v1 Def 1.7 (Wit_k in the rank- $(k + 1)$ state, line 367): PP/PPI existentials are removed from the witness menu and discharged by reachability on the parent graph instead.
- v1 Thm 1.20 (decidability sketch, line 681–691): the proof sketch is dropped and replaced by a citation to v2 completeness extraction [6], which carries out the model-to-quotient construction in full.

1.2 What v2 does not change

This note does not introduce any of the following:

- No Büchi automata, ω -words, or parity acceptance: PP/PPI eventualities are handled by ordinary finite-graph reachability, not by ω -language theory.
- No new model class: split-forest semantics, Hintikka types, RCC5 atomic relations, and strong-EQ models are exactly as in v1.
- No new descriptor algebra: the local coherence axioms (C1)–(C5) are the v1 axioms, applied to the same rank- k descriptors. The witness-conflation defect that GPT-5.5 identified in the v1 Thm 1.20 sketch is not a defect in the axioms themselves; it is a defect in how the v1 sketch *extracted* the axiom-satisfying tables from a model. That extraction is carried out properly in v2 completeness extraction [6, §6, §9].

1.3 Why a v2 paper rather than an in-place edit

GPT-5.4 wrote the v1 paper in April 2026, before GPT-5.5’s review appeared. Editing it in place would mix attribution: the reinterpreted definitions and the replaced Thm 1.20 sketch are Claude’s, not GPT-5.4’s. v1 is therefore retained unchanged in the repository, marked as superseded; v2 is this separate Claude note. The arrangement parallels the v1 [5]/v2 [6] split for the companion completeness-extraction paper.

2 The split-tree presentation, reinterpreted

2.1 The defect in v1 Def 1.5

v1 Def 1.5 defines a weak-EQ split-tree presentation as a rooted tree T with an *ambient root* ϱ_T above the distinguished evaluation node r_* . Tree edges are read as “immediate PPI edges downward and hence PP upward,” and all indirect PP/PPI edges are recovered from the ancestor relation. This presupposes that every relevant model has a PP-maximal element above the evaluation point, in the same sense as a Hasse-cover diagram of a well-founded order.

GPT-5.5’s review [2] identifies a counterexample:

$$C_{\uparrow} := \exists \text{PP}.\top \sqcap \forall \text{PP}.\exists \text{PP}.\top.$$

Under the v1 orientation (PP-witness = strict superpart), this concept is satisfiable in any model whose domain contains a strictly increasing PP-chain

$$a_0 \text{ PP } a_1 \text{ PP } a_2 \text{ PP } \dots$$

with $C_\uparrow$ realised at a_0 . No element of this domain is PP-maximal, so no ambient root ϱ_T exists. Yet the concept is satisfiable. Therefore v1 Def 1.5 is too restrictive: it does not admit a presentation for every satisfiable concept.

The defect is structural rather than cosmetic. It is what GPT-5.5 calls the *rooted Hasse* assumption [2, 4].

2.2 Reinterpretation: generated witness-cover presentations

The fix preserves the tree-shaped containment backbone and the PPI-down/PP-up reading, but drops the ambient-root requirement and reinterprets tree edges as *generated* witness-cover edges relative to a finite selected witness set. The construction is carried out in detail in v2 completeness extraction [6, §4–§5]; we summarise it here.

Definition 2.1 (Generated weak-EQ split-tree presentation; v2 reinterpretation of v1 Def 1.5). A *generated weak-EQ split-tree presentation* of a candidate model of C_0 consists of:

- (i) a tree T of *occurrences* with a distinguished *evaluation node* $r_* \in T$ (no ambient root is required);
- (ii) a partial parent function $\text{par} : T \rightarrow T$, allowing $\text{par}(u) = u$ at the root of an upward chain (*parent self-loop*);
- (iii) a map $\text{orig} : T \rightarrow D$ onto an underlying generated submodel D (defined below);
- (iv) a local type assignment $\lambda : T \rightarrow \text{Tp}(C_0)$;

such that:

- tree edges are interpreted as immediate PPI edges downward and hence PP upward, *relative to the finite witness set* $\text{orig}(T)$ *used in the construction*; they are not claimed to be Hasse covers of any ambient PP-order;
- if a node of D must appear under multiple distinct parents, it may be represented by several occurrences in T (split copies);
- two occurrences are in relation EQ exactly when they are split copies of the same element of D that satisfy the typed-EQ congruence of v1 Def 1.16.

A parent self-loop $\text{par}(u) = u$ records that u realises an upward PP-existential whose witness has the same rank- d profile as u . The unfolding (Section 2.4) expands such a self-loop into a fresh occurrence at every level.

Remark 2.2 (Why parent self-loops absorb $C_\uparrow$). For $C_\uparrow$, every element of the witnessing chain $a_0 \text{ PP } a_1 \text{ PP } \dots$ has the same rank- d profile (local type $\{C_\uparrow, \exists \text{PP}.\top, \forall \text{PP}.\exists \text{PP}.\top, \dots\}$, plus the $\text{Par}_d/\text{Child}_d$ multiplicities which are stationary along the chain). The finite quotient therefore identifies all a_i at a single rank- d state σ and records the parent function as $\text{par}_Q(\sigma) = \sigma$.

Unfolding this self-loop produces the infinite chain $a_0\text{PP}a_1\text{PP}\dots$ as required; $C_\uparrow$'s satisfiability is thus represented by a quotient of size 1 (one rank- d state, one parent edge, a self-loop), with no ambient root.

Remark 2.3 (Generated cover edges, in one paragraph). The phrase “generated witness-cover edge” [4, 6] means the following. Fix a witness function w that, for each occurrence u and each $\exists R.D \in \lambda(u)$, selects a model element witnessing the existential. Let A be the finite set of occurrences built so far. A generated PPI-cover edge $u \rightarrow v$ is added when $\rho(\text{orig}(u), \text{orig}(v)) = \text{PPI}$ and no other occurrence $z \in A$ has $\rho(\text{orig}(u), \text{orig}(z)) = \text{PPI}$ and $\rho(\text{orig}(z), \text{orig}(v)) = \text{PPI}$. This is well-defined regardless of whether the ambient PP-order is dense or has Hasse covers: the cover is taken *relative to* A , not in the model. As A grows, new occurrences may slot between previously adjacent ones; the construction proceeds inductively and is carried out explicitly in [6, Construction 5.1].

2.3 Compatibility with v1 soundness theorems

v1 Theorems 1.17–1.19 are statements about *abstract valid rank- d quotients*: given a valid quotient Q in the sense of v1 Def 1.10, an unfolding $U(Q)$ is built, the disjunctive network of every finite prefix has non-empty Need $_R$ -filtered domains, canonical refinements yield a globally consistent atomic labelling, and the resulting weak-EQ model is then quotiented to a strong-EQ model satisfying C_0 . None of these proofs inspect whether the quotient came from a Hasse construction or a generated-cover construction. They only use:

- finiteness of S (rank- d states);
- finiteness of the disjunctive network alphabet;
- the local coherence axioms (C1)–(C5);
- the parent slot Par_d and child multiplicities Child_d as components of each state.

Allowing $\text{par}_Q(\sigma) = \sigma$ does not break any of these proofs: the unfolding step (v1 Def 1.11) constructs a fresh occurrence at each predecessor level, and a self-loop simply means that every level shares the same state. König's lemma in v1 Lemma 1.13 still produces an infinite branch when one is needed, and the canonical-refinements argument of v1 Lemma 1.16 treats every prefix uniformly.

Lemma 2.4 (v1 soundness chain is preserved). *Replacing v1 Def 1.5 with the generated split-tree presentation of Definition 2.1 (parent self-loops permitted, no ambient root) leaves v1 Theorems 1.17, 1.18, and 1.19 valid without modification.*

Proof sketch. Each of these theorems takes a valid finite rank- d quotient Q as input and constructs an RCC5 model. The proofs use the finite-state quotient structure $(S, \text{par}_Q, \text{Child}_Q, \text{Iface}_Q)$, not Def 1.5 directly. The only place Def 1.5 enters is when the unfolding $U(Q)$ is described as a tree of occurrences with PPI-down edges. With self-loops permitted, $U(Q)$ becomes the regular unfolding of the parent graph: each $\text{par}_Q(\sigma) = \sigma$ self-loop expands into an infinite branch of fresh occurrences with state σ . The arc-consistency theorem (v1 Thm 1.12) iterates over finite prefixes, all of which remain finite (each parent step adds one occurrence). König's lemma (v1

Lemma 1.13) extracts a globally consistent atomic labelling from compatibility on every finite prefix. The canonical-refinements / typed-EQ-congruence steps (v1 Lemmas 1.15 and 1.16) operate on the resulting infinite labelling. Strong-EQ realisation (v1 Thm 1.19) quotients by the typed-EQ congruence. None of these steps need an ambient root. $\square$

2.4 Unfolding of a self-loop

For completeness we record what the unfolding of a parent self-loop looks like, since the v1 unfolding (Def 1.11) only covered the rooted-tree case.

Definition 2.5 (Unfolding $U(Q)$, v2). Let $Q = (S, \text{par}_Q, \text{Child}_Q, \text{Iface}_Q, \sigma_*)$ be a valid finite rank- d quotient with distinguished evaluation state $\sigma_* \in S$. The *unfolding* $U(Q)$ is the (possibly infinite) tree of occurrences defined coinductively:

1. create a root occurrence r_* with $\text{st}(r_*) = \sigma_*$;
2. for each occurrence u with $\text{st}(u) = \sigma$ and each child slot $\sigma' \in \text{Child}_Q(\sigma)$ (with multiplicity 1 or +), create one or more child occurrences with $\text{st} = \sigma'$;
3. for each occurrence u with $\text{st}(u) = \sigma$, if $\text{par}_Q(\sigma) = \sigma'$ for some $\sigma' \neq \sigma$, create a fresh parent occurrence $u^\uparrow$ with $\text{st}(u^\uparrow) = \sigma'$ and set $\text{par}_{U(Q)}(u) = u^\uparrow$;
4. if $\text{par}_Q(\sigma) = \sigma$ (self-loop), create a fresh parent occurrence $u^\uparrow$ with $\text{st}(u^\uparrow) = \sigma$ and set $\text{par}_{U(Q)}(u) = u^\uparrow$. The same rule applies to $u^\uparrow$, producing an infinite chain of fresh occurrences all in state σ .

Remark 2.6 (Self-loops are blocking, not equality). A parent self-loop $\text{par}_Q(\sigma) = \sigma$ records that the rank- d *profile* repeats; it does *not* identify the two occurrences as EQ-mates. The unfolding produces fresh occurrences at every level. This is the distinction stressed in [4]: blocking equivalence and typed-EQ congruence are different. Only the latter justifies a quotient of the unfolded model.

3 PP/PPI eventualities by reachability discharge

3.1 The defect in v1 Def 1.7

v1 Def 1.7 (rank- $(k+1)$ state, line 367) records, as part of each state σ , a finite *witness menu* $\text{Wit}_k(\sigma)$ recording, for every existential $\exists R.D \in \lambda(\sigma)$, “one designated target color and target relation domain.” The patched cover-tree tableau note [7] also puts PP/PPI existentials in the witness menu via **ReqUp** and **ReqDown** slots. By contrast, the v1 completeness extraction [5] states (lines 399–401) that $\exists \text{PP}.D$ and $\exists \text{PPI}.D$ are handled by the tree structure and *do not enter the witness menu*.

GPT-5.5 [2] flagged this inconsistency and pointed out a real obstacle: a designated target color does not, by itself, guarantee that the existential is realisable in the unfolded model. For transitive roles, the witness for $\exists \text{PP}.D$ at σ might lie arbitrarily far up the parent chain — in particular, beyond any finite parent slot.

3.2 The fix: reachability over the parent graph

The v2 completeness extraction [6, §11] discharges PP/PPI eventualities by ordinary reachability over the finite parent graph (S, par_Q) .

Definition 3.1 (Parent reachability; cf. [6] Def 11.1). For $\sigma, \sigma' \in S$, write $\sigma \rightsquigarrow_{\text{PP}} \sigma'$ when there exist $\sigma_0, \sigma_1, \dots, \sigma_n$ in S with $\sigma_0 = \sigma$, $\sigma_n = \sigma'$, $n \geq 1$, and $\text{par}_Q(\sigma_i) = \sigma_{i+1}$ for $0 \leq i < n$. A self-loop $\text{par}_Q(\sigma) = \sigma$ counts as $\sigma \rightsquigarrow_{\text{PP}} \sigma$. Symmetrically for $\rightsquigarrow_{\text{PPI}}$ via the inverse parent function (i.e., on child edges).

Definition 3.2 (Reachability discharge condition (v2 axiom)). A valid rank- d quotient Q must satisfy: for every $\sigma \in S$ and every $\exists R.D \in \lambda_Q(\sigma)$ with $R \in \{\text{PP}, \text{PPI}\}$, there exists $\sigma' \in S$ with $\sigma \rightsquigarrow_R \sigma'$ and $D \in \lambda_Q(\sigma')$.

Remark 3.3 (v2 amendment to v1 Def 1.10). v1 Def 1.10 (valid finite rank- d quotient) is amended as follows: replace the witness-menu condition for PP/PPI existentials by Definition 3.2. Existentials with $R \in \{\text{DR}, \text{PO}\}$ continue to be handled by the witness menu Wit_k as in v1 Def 1.7. The witness menu therefore retains its Need_R -filtering side conditions for DR and PO; it simply no longer carries any ReqUp/ReqDown slots.

Remark 3.4 (This is finite-graph reachability, not Büchi). Definition 3.1 is reachability in a finite directed graph, decidable in $O(|S| + |E_Q|)$ per eventuality by breadth-first search. There is no ω -language theory and no acceptance condition. A self-loop $\text{par}_Q(\sigma) = \sigma$ certifies that the parent chain in the unfolding is infinite and remains in the strongly connected component of σ ; if some σ' in that SCC satisfies $D \in \lambda_Q(\sigma')$, the eventuality is discharged. See [6, §11] and [6, Remark 11.5] for the comparison with GPT-5.5's automata-theoretic formulation [3, Def 6.4].

3.3 Compatibility with v1 soundness theorems

The amendment to v1 Def 1.10 changes which quotients count as valid; it does not change what valid quotients unfold to. v1 Thms 1.17–1.19 take a valid quotient as input and proceed purely in terms of S , par_Q , Child_Q , and the disjunctive networks of finite prefixes. None of those proofs relies on Wit_k carrying ReqUp/ReqDown slots. The arc-consistency theorem (v1 Thm 1.12) only needs Need_R -filtering for $R \in \{\text{DR}, \text{PO}\}$, which is the part of Wit_k that v2 retains.

Lemma 3.5 (Soundness chain unchanged). *Replacing the v1 witness-menu treatment of PP/PPI existentials by reachability discharge (Definition 3.2) leaves v1 Theorems 1.17–1.19 valid without modification.*

Proof sketch. A valid rank- d quotient under the v2 axioms is also valid under the v1 axioms restricted to the part actually used by Thms 1.17–1.19 (witness menus for DR/PO, plus all of (C1)–(C5), Need_R -filtering, and the parent/child slots). PP/PPI existentials are realised in the unfolding by a finite parent-chain step: if $\sigma \rightsquigarrow_{\text{PP}} \sigma'$ in Q with $D \in \lambda_Q(\sigma')$, the unfolding contains a finite branch $u_0, \dots, u_n$ with $\text{st}(u_0) = \sigma$, $\text{st}(u_n) = \sigma'$, and $D \in \lambda(u_n)$. By v1 Lemma 1.4 (closure under modal depth), the universal companions of $\exists \text{PP}.D$ propagate correctly along this chain, and the local-type Hintikka conditions are preserved. $\square$

4 Decidability via the v2 chain

4.1 Replacement for v1 Thm 1.20

Theorem 4.1 (Decidability of $\mathcal{ALCI}_{RCC5}$ concept satisfiability; v2). *Concept satisfiability in $\mathcal{ALCI}_{RCC5}$ under strong-EQ semantics is decidable.*

Proof. Combine three results.

Soundness (valid quotient $\rightarrow$ model). By v1 [1, Thms 1.17, 1.18, 1.19], every valid finite rank- d quotient Q unfolds to an arc-consistent disjunctive network whose König-extracted canonical refinement quotients to a strong-EQ model of C_0 . Lemmas 2.4 and 3.5 establish that the v2 reinterpretation of Def 1.5 (parent self-loops, generated covers) and the v2 amendment of Def 1.10 (PP/PPI-by-reachability-discharge) leave these theorems valid.

Completeness (model $\rightarrow$ valid quotient). By v2 completeness extraction [6, Thm 12.1], every pointed strong-EQ model $(\mathcal{I}, a_0) \models C_0$ extracts to a valid finite rank- d quotient Q whose distinguished evaluation state contains C_0 . The construction uses witness-generated submodels [6, §3], generated cover edges [6, §4], support-closed mosaics [6, §6], and reachability discharge [6, §11].

Effective enumeration. By v1 finite-index lemma (Thm 1.10), the set Σ_d of rank- d states is finite; hence finitely many candidate quotients exist up to isomorphism. The v2 validity conditions (v1 Def 1.10 with Def 1.7 and Def 1.5 replaced as above) are decidable on a finite quotient: each of (C1)–(C5) is a finite local check; reachability discharge is finite-graph reachability; the witness menu’s Need_R-filtering for DR/PO existentials is a finite check on the disjunctive network alphabet.

Therefore: enumerate finite quotients up to isomorphism, check v2 validity, accept iff a valid quotient with $C_0 \in \lambda(\sigma_*)$ is found. By soundness, acceptance implies C_0 is satisfiable. By completeness, satisfiability of C_0 implies a valid quotient exists. The procedure is decidable. $\square$

4.2 What is not claimed

- No complexity bound. The quotient enumeration is non-elementary in $|C_0|$ on its face. An EXPTIME upper bound (matching the $\mathcal{ALCI}$ lower bound) would require a tighter analysis of $|\Sigma_d|$, which is open. v1 simply asserted finiteness of Σ_d without a quantitative bound; v2 does the same.
- No claim about $\mathcal{ALCI}_{RCC8}$. Both v1 and v2 only treat the RCC5 case. Extension to RCC8 (which also has the patchwork property of Renz & Nebel [9]) is expected to carry through by symmetric reasoning, but has not been written out.
- No formal equivalence with the cover-tree tableau implementation [8]. The implementation is an algorithmic realisation that is empirically validated on 911 + 400 cross-validated test concepts and 12.2M enumerated models, but the formal equivalence between its CT1–CT5 checks and v2 quotient validity is a separate open claim.

5 Comparison with v1 (line-level)

For ease of cross-reference with v1, this section lists the specific edits, by v1 line number, that v2 induces.

v1 location	v1 reading	v2 reinterpretation
Def 1.5 (line 161)	“rooted tree T with ambient root ϱ_T ”	no ambient root; par may self-loop at root (Def 2.1)
Def 1.5 (line 167)	“tree edges are immediate PPI edges”	generated witness-cover edges relative to $A = \text{orig}(T)$ (Remark 2.3)
Def 1.7 (line 367)	Wit_k records target color for each existential, including PP/PPI	Wit_k retains DR/PO slots only; PP/PPI discharged by reachability (Def 3.2)
Thm 1.20 (line 681–691)	decidability proof sketch with ambient-root choice and “least support-closed tables” assertion	proof replaced by Theorem 4.1; model-to-quotient delegated to [6, Thm 12.1]
Remark following Thm 1.20 (line 693–695)	“the only compressed part is the extraction of support-closed tables”	extraction is now explicit [6, §6, §9]

All other line-by-line content of v1 stands.

6 Conclusion

GPT-5.4’s v1 sibling-interface paper [1] contains a soundness chain (Thms 1.17–1.19) for the split-forest decidability programme that GPT-5.5’s review [2] explicitly listed as salvageable, alongside two structural defects in adjacent material: an ambient-root assumption in Def 1.5 that fails on $C_\uparrow$, and a Thm 1.20 proof sketch that asserts but does not construct support-closed tables and conflates witnesses across distinct occurrences when discharging the C5 axiom. v2 replaces Def 1.5 with a generated witness-cover presentation (parent self-loops permitted), replaces the witness-menu treatment of PP/PPI existentials by reachability discharge on the finite parent graph, and replaces the Thm 1.20 sketch with a citation to v2 completeness extraction [6]. The soundness chain of Thms 1.17–1.19 is preserved verbatim. Combined with v2 completeness extraction, this yields a decidability theorem for $\mathcal{ALCI}_{RCC5}$ concept satisfiability under strong-EQ semantics, without Büchi machinery or ω -acceptance.

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